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Word-Level ASL Recognition from Hand Landmarks

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Agenda Overview

01 Introduction

02 Goals

03 Work Plan

04 Architecture

05 Experimental Approach

06 Results

07 Ethical Analysis

08 Conclusion

Introduction

THE PROBLEM

Sign languages are **natural languages** with their own visual-spatial structure, not simple gesture codes. Their meaning emerges from **hand shape, motion, orientation, location, and timing.**

WHY IS IT HARD?

The same sign varies across **signers, viewpoints, execution speeds, and recording conditions**, limiting the reach of every accessibility tool that depends on it.



Goals

MAIN GOAL

Design, implement, and experimentally validate a research prototype for word-level **American Sign Language (ASL)** recognition from short video clips, using hand-landmark representations and temporal neural models.

RESEARCH HYPOTHESIS

*Can **hand-landmark** sequences provide enough information for a temporal model to classify short ASL video clips into **word glosses**?*



Specific Goals

SO1
Analyse the technical and scientific background of SLR.

S02
Define a landmark-based data representation for ASL clips.

S03
Compare candidate temporal architectures under matched conditions.

S04
Validate the winner model with statistically meaningful protocols.

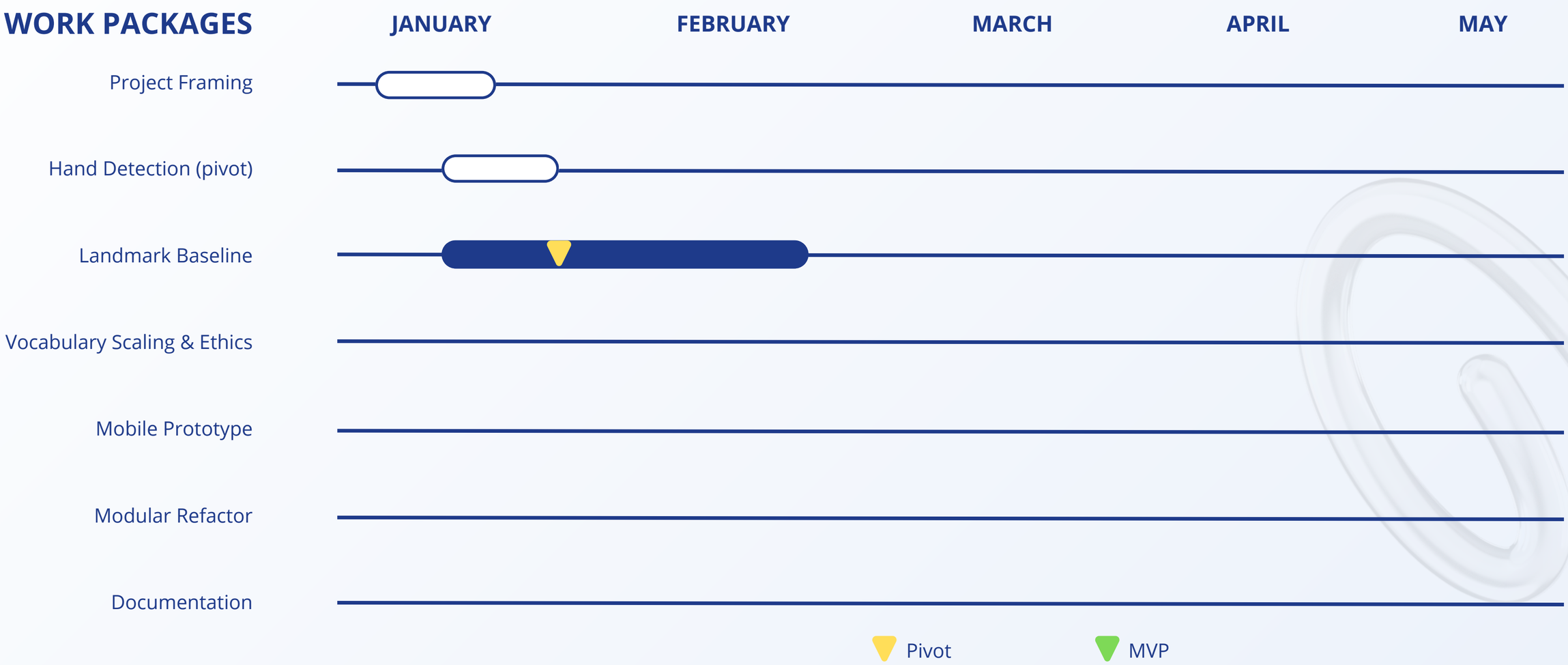
S05
Verify generalisation through cross-dataset and vocabulary scaling.

S06
Build a runtime prototype and evaluate the ethical implications.

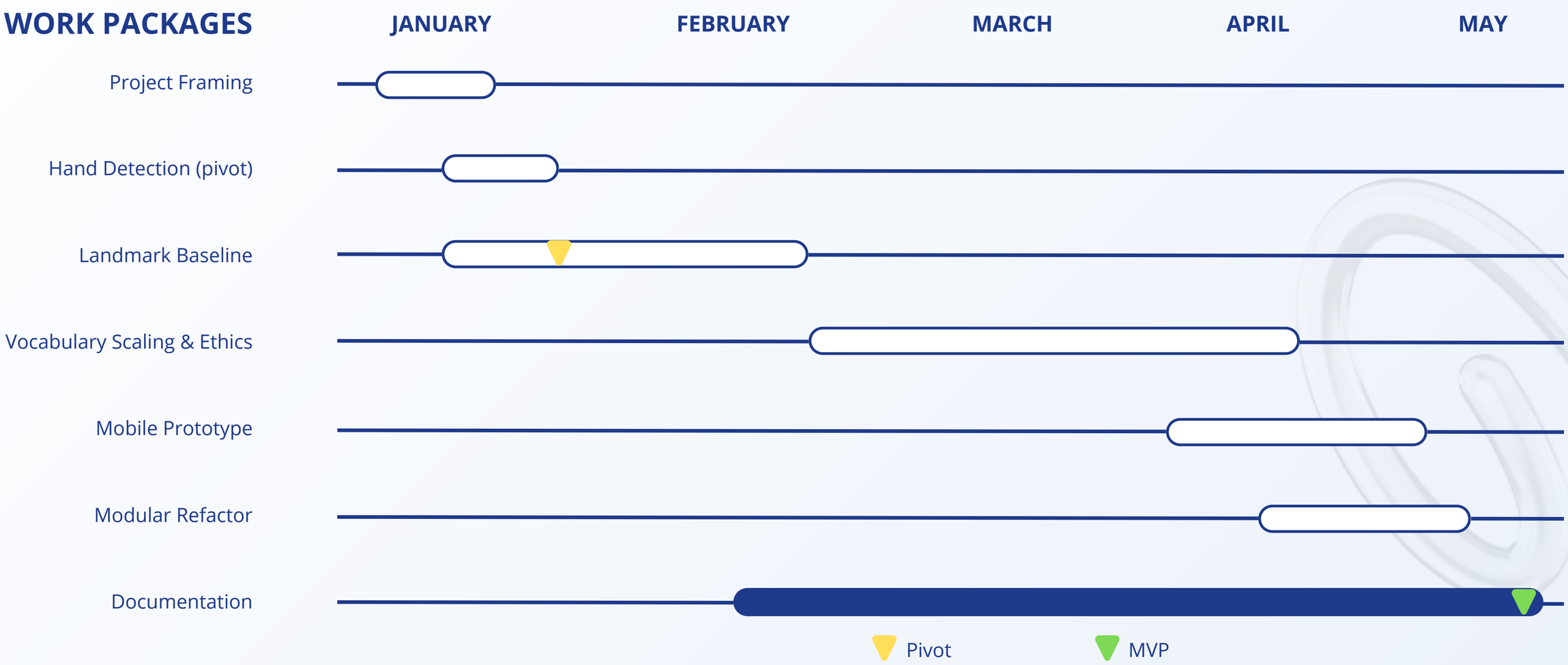
S07 EXTRA BEYOND RESEARCH SCOPE

Deploy a fully functional mobile prototype with real-time ASL recognition.

Work Plan


WORK PACKAGES
JANUARY
FEBRUARY
MARCH
APRIL
MAY
Project Framing
Hand Detection (pivot)
Landmark Baseline
Vocabulary Scaling & Ethics
Mobile Prototype
Modular Refactor
Documentation
Pivot
MVP

Work Plan

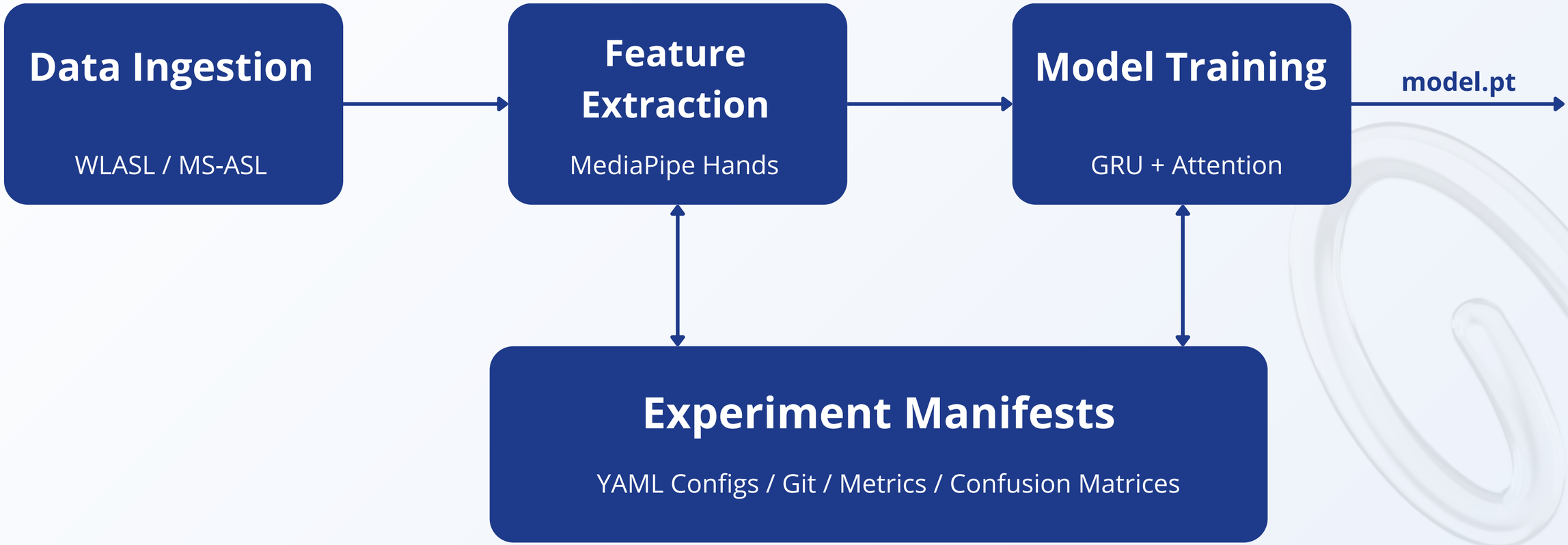

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Architecture

Training Architecture

Inference Architecture

Training Architecture



Inference Architecture

Mobile Client

Expo &
React Native



FastAPI Server

Uvicorn



MediaPipe Hands

21 key
points/frame



GRU + Attention

Model

Top-K Prediction

“no” 0.87 / “before” 0.06 / “cool” 0.03

RESEARCH QUESTION

*Can **hand-landmark** sequences provide enough information for a temporal model to classify short ASL video clips into **word glosses**?*

EXPERIMENTAL APPROACH

01
DATASET PREPARATION

MediaPipe 21 keypoints

02
ARCHITECTURE SELECTION

GRU + Attention

03
SEQUENCE SWEEP

64 frames / stability

04
WLASL30 VALIDATION

5-Fold CV + multi-seed

05
CROSS DATASET

MS-ASL50 generalisation

06
VOCABULARY SCALING

WLASL 30 → 1500

Results

EXPERIMENTAL RESULTS - DETAILED

EXPERIMENT / CONDITION	SETUP	TOP-1 (%)	TOP-5 (%)	NOTE
02 ARCHITECTURE SELECTION (WLASL30, multi-seed)				
LSTM	landmark-based	41.18	67.65	baseline recurrent
GRU	landmark-based	38.24	76.47	plain recurrent
GRU + Features	hand-engineered	47.06	79.41	enriched input
GRU + Attention	temporal attention	58.82	91.18	WINNER
Transformer	self-attention	50.00	73.53	underfits low-data
04 STATISTICAL VALIDATION (WLASL30 Winner)				
5-Fold CV	WLASL100 → 30	63.85 ± 4.50	86.46 ± 2.04	headline number
Multi-seed	fixed split / 3 seeds	58.82 ± 2.94	78.43 ± 4.49	stability check
05 CROSS-DATASET GENERALISATION (no architectural changes)				
WLASL30	30 classes / 34 test	58.82 ± 2.94	78.43 ± 4.49	baseline
MS-ASL50	50 classes / 236 test	70.20 ± 2.89	90.54 ± 0.89	σ ≈ WLASL30 σ
06 VOCABULARY SCALING (WLASL, single seed)				
100 classes	795 train	44.70	70.45	−14 pp vs 30
300 classes	1,974 train	28.61	59.72	log-linear regime
700 classes	3,879 train	19.57	42.66	instability threshold
1000 classes	5,122 train	11.19	31.24	−47 pp vs 30
1500 classes	6,864 train 4.6 vid/class	9.76	28.23	data starvation

Experiments 01 (preparation) and 03 (sequence sweep, σ=0.93pp selected 64 frames) summarised in setup.. ★ key results

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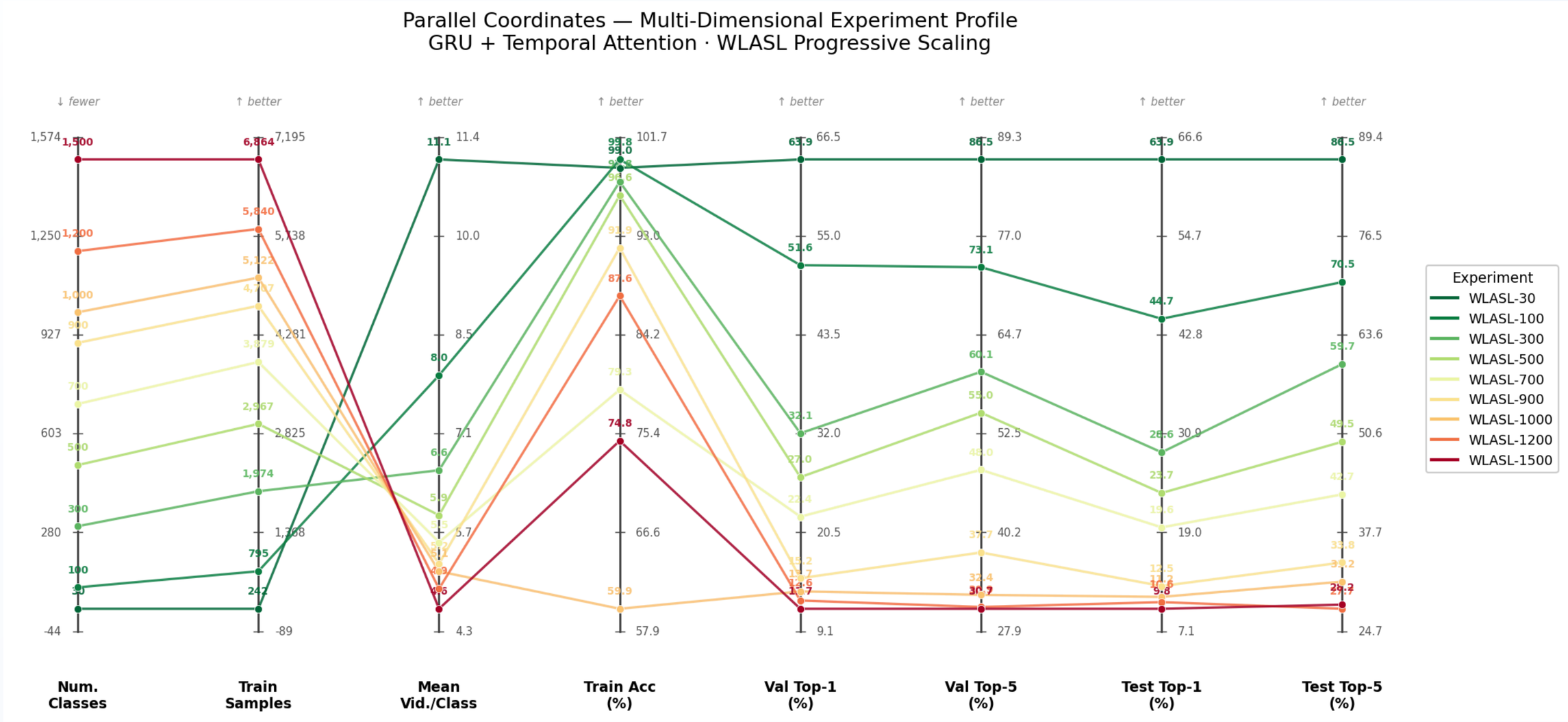
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Results


Parallel Coordinates — Multi-Dimensional Experiment Profile
GRU + Temporal Attention · WLASL Progressive Scaling
↓ fewer
↑ better
↑ better
↑ better
↑ better
↑ better
↑ better
↑ better
Num. Classes
Train Samples
Mean Vid./Class
Train Acc (%)
Val Top-1 (%)
Val Top-5 (%)
Test Top-1 (%)
Test Top-5 (%)
Experiment
WLASL-30
WLASL-100
WLASL-300
WLASL-500
WLASL-700
WLASL-900
WLASL-1000
WLASL-1200
WLASL-1500

USER STUDY GOOGLE FORMS

SignToText real users feedback

N=15 participants | 5min survey | Likert + open-ended

Qualitative Validation

USABILITY DIMENSIONS (1–5 LIKERT)



TOP-VALUED FEATURES (% MENTIONS)

- Top-K alternatives with confidence 87%
- Automatic text-to-speech 73%
- Multiple model selection 60%

NET PROMOTER SCORE

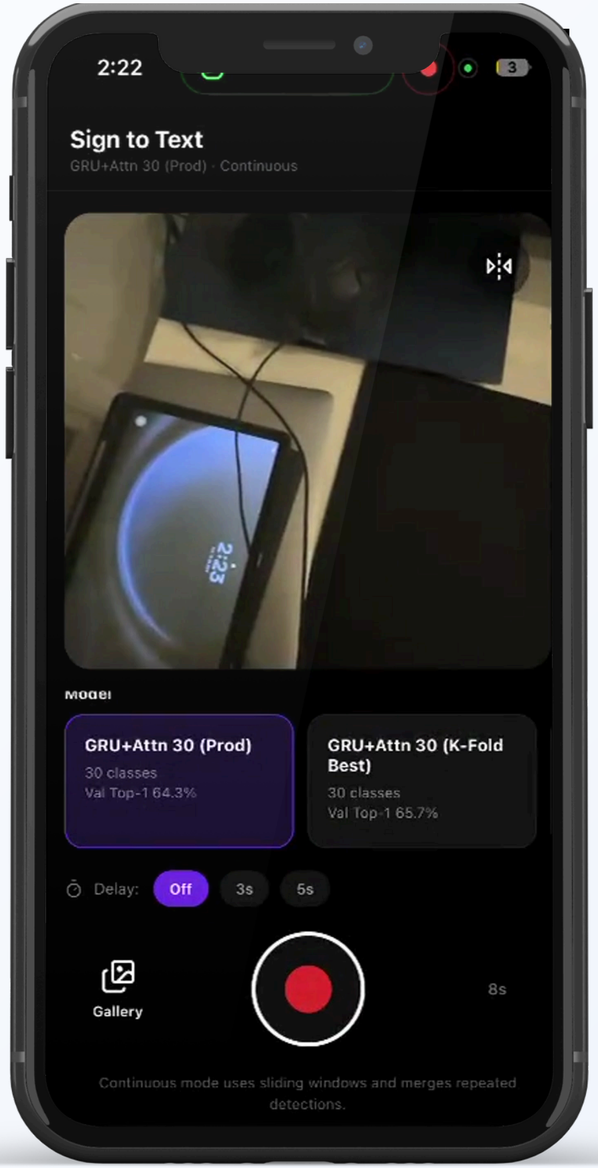
+40

(industry good: +30 to +50)

MAIN FRICTION

Attention map and processing stats confused users without context

DEMO


2:22
3
Sign to Text
GRU+Attn 30 (Prod) · Continuous
Model
GRU+Attn 30 (Prod)
30 classes
Val Top-1 64.3%
GRU+Attn 30 (K-Fold Best)
30 classes
Val Top-1 65.7%
Delay:
Off
3s
5s
Gallery
8s
Continuous mode uses sliding windows and merges repeated detections.

FRAMEWORK

Google’s AI Principles

26 metrics | 7 principles | scored 0 / 5 / 10

Ethical Analysis

SCORES BY PRINCIPLE



OVERALL

7.6 / 10

WEAKEST POINT

Privacy in inference: raw frames on server

Key insight: MediaPipe landmarks as input originally a technical decision, propagate positive effects across P2 (bias), P5 (privacy) and P7 (misuse). One choice, three ethical wins.

Conclusions

ANSWERING THE RESEARCH QUESTION

*"Can **hand-landmark** sequences provide enough information for a temporal model to classify short ASL video clips into **word glosses**?"*

YES

Within the tested vocabulary scale, with reproducible stability across two ASL benchmarks ethically grounded.

GENERAL GOAL

Design, implement, and experimentally validate a research prototype for word-level ASL recognition from short video clips, using hand-landmark representations and temporal neural models.

Specific Goals

SO1 ✓
Technical background analysed
State of the Art chapter

S02 ✓
Landmark representation defined
~69% WLASL ~84% MS-ASL50

S03 ✓
5 architectures compared
GRU + Attention winner

S04 ✓
Statistical validation
63.85% ± 4.50% 5-Fold CV

S05 ✓
Cross-dataset + scaling
MS-ASL50 70.20% WLASL 30→1500

S06 ✓
Prototype + ethics
Mobile + FastAPI 7.6/10 ethics

S07 ✓ EXTRA DELIVERED
Real-time mobile app deployed
Expo + React Native & FastAPI backend / production-ready

FROM THIS RESEARCH TO

three frontiers worth pursuing

Next Steps

01

Beyond word-level & continuous translation

Move from isolated glosses to full ASL sentences, the open frontier of sign language recognition.

TECHNICAL

02

Cross-lingual sign transfer

Transfer the landmark-based pipeline from ASL to LSE, BSL, and beyond, since hand geometry is universal, vocabularies are not.

SCIENTIFIC

03

Privacy-first edge deployment

Move the entire pipeline on-device so only landmarks ever leave the phone, closing the privacy gap identified in the ethical analysis.

APPLIED

Author:
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Thank You